
CURRICULUM VITAE

Alexandra Veliche Hostetler

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Education

University of Michigan, Ann Arbor, MI

PhD in Computer Science and Engineering

Graduation Date: August 2025

Advisor: Mahdi Cheraghchi

Northeastern University, Boston, MA

Bachelor of Science in Mathematics, Minor in Computer Science

Graduation Date: May 2020

Summa Cum Laude

Research Interests

I am broadly interested in theoretical computer science and its mathematical foundations. I am especially interested in the foundations of cryptography, computational complexity, and the connections between error-correcting codes and geometric lattices.

Publications

[BHJ,2026] - Jean-François Biasse, Alexandra Veliche Hostetler, and Anuvrat Jainundarwal. *On the Hull Attacks Against Construction A Lattices*. Selected Areas in Cryptography (SAC) 2026, Ottawa, Ontario, Canada.

[PV,2025] - Chris Peikert and Alexandra Veliche Hostetler. *List Decoding Reed–Solomon Codes in the Lee, Euclidean, and Other Metrics*. 17th Innovations in Theoretical Computer Science Conference (ITCS), 2026.

[CSV,2025] - Mahdi Cheraghchi, Nikhil Shagrirhaya, and Alexandra Veliche. *Reductions Between Code Equivalence Problems*. 2025 IEEE International Symposium on Information Theory (ISIT), Ann Arbor, MI, U.S.A., pages 1–6, 2025.

[ALV,2024] – Divesh Aggarwal, Jin Ming Leong, and Alexandra Veliche. *Worst-Case to Average-Case Hardness of LWE: An Alternative Perspective*. Theory of Cryptography, pages 308–336, Cham, 2025. Springer Nature Switzerland.

[CDRV,2022] – Mahdi Cheraghchi, Joseph Downs, João Ribeiro, Alexandra Veliche, *Mean-Based Trace Reconstruction over Oblivious Synchronization Channels*. IEEE Transactions on Information Theory, 68(7):4272-4281, 2022.

[CDRV,2021] – Mahdi Cheraghchi, Joseph Downs, João Ribeiro, Alexandra Veliche, *Mean-Based Trace Reconstruction over Practically any Replication-Insertion Channel*. 2021 IEEE International Symposium on Information Theory (ISIT), Melbourne, Australia, pages 2459-2464, 2021.

Invited Talks

ACCESS Seminar – online

June 2025

“List Decoding Reed-Solomon Codes in the Lee, Euclidean, and Other Metrics”.

Guest Lecture, MATH 4757: Introduction to Cryptography, Northeastern University – Boston, MA

December 2025

“Cryptography based on the Code Equivalence and Lattice Isomorphism Problems”.

Theory of Computer Science Seminar, Purdue University – West Lafayette, IN

September 2024

“Worst-case to Average-case Hardness of LWE: An Alternative Perspective”.

Computing Theory Seminar, National University of Singapore – Singapore

May 2023

“Mean-Based Trace Reconstruction over Oblivious Synchronisation Channels”.

Teaching

Mathematics Department of University of South Florida

Tampa, FL

Instructor for Calculus 3 (MAC 2313)

Fall 2026

CodeBreakHERs Summer Camp

Summer 2026

Instructor for Modern Cryptography (MAD 4472)

Fall 2025

CSE Department of University of Michigan

Ann Arbor, MI

Graduate Student Instructor for Discrete Mathematics (EECS 203)

Spring 2025

Graduate Student Instructor for Introduction to Cryptography (EECS 475)

Fall 2023

Graduate Student Instructor for Advanced Cryptography (EECS 575)

Fall 2022

Graduate Student Instructor for Introduction to Algorithms (EECS 477)

Fall 2021

New Horizons Summer School

Online

Volunteer Teaching Assistant for Cryptography

June 2021

Computer Science Department of Northeastern University

Boston, MA

Teaching Assistant for Cryptography (CS 4770)

January – April 2020

St. Herman of Alaska Christian School

Allston, MA

Volunteer Teacher’s Assistant for Middle School Geometry Class

Fall 2016 – Fall 2019

Mathematics Department of Northeastern University

Boston, MA

Mathematics Tutor

May 2017 – April 2019

Internships & Research Visits

National University of Singapore, Centre for Quantum Technologies <i>Research Visitor / Intern</i>	Singapore Summer 2023
Center for Communications Research, Princeton <i>Researcher</i>	Princeton, NJ Summer 2022
U.S.A. Department of Defense <i>Director's Summer Program Participant</i>	MD Summer 2019
Cengage Learning <i>Cybersecurity Co-op</i>	Boston, MA Spring – Summer 2018

Honors & Awards

Rackham Conference Travel Grant Awarded to support travel expenses for conference presentation at TCC.	September 2024
Honorable Mention Award for Teaching Recognized for commitment as a Graduate Student Instructor.	May 2022
ISIT 2021 Four Minutes, Two Techniques Student Challenge Winner Awarded to top 3 teams for a collaborative 4-minute video explaining 2 techniques in coding theory.	July 2021
PEAK Shout-It-Out Award Travel award to present a poster at the NCUWM in February 2020.	January 2020
Churchill Scholarship Nominee Nominated by Northeastern University for a fully sponsored master's degree at Cambridge University, U.K.	November 2019
National Merit Scholarship Merit-based scholarship sponsored by Liberty Mutual, awarded per semester.	Fall 2016 – Spring 2020

Service

CSE Department of University of Michigan <i>Theory Lab Retreat</i> Collaborated with two other graduate students to organize and lead a weekend retreat for the graduate students in our theory group to promote interdisciplinary collaboration and improve the lab culture.	Ann Arbor, MI Fall 2024
CSE Department of University of Michigan <i>Theory Lunch Seminar Organizer</i> Collaborated with other graduate students to organize a joint faculty and student lunch seminar, as well as a student-only seminar for the theory group.	Ann Arbor, MI Fall 2021 – Spring 2022, Fall 2024
